We consider the **Unsplit Compartments–Split Tanks** variant. A vehicle can perform at most one route in each period. Every used compartment contains one product and, because compartments are unsplit, its **full content is delivered to one customer tank**. A customer tank may nevertheless receive deliveries from more than one vehicle in the same period; this is the meaning of *split tanks* in Coelho and Laporte.

The formulation below uses directed arcs. Consequently, every routing variable is binary. A route containing only one customer is represented by two distinct arcs, $0 \to i$ and $i \to 0$. This avoids the special value 2 required by the undirected depot-edge variable in the original paper.

A.2 Sets and indices

- $\mathcal{V} = \{0,1, \dots, n\}$: set of nodes.
- 0: supplier/depot.
- $\mathcal{V}' = \{1, \dots, n\}$: set of customers.
- $\mathcal{M}$: set of products, indexed by m.
- $\mathcal{K}$: set of vehicles, indexed by k.
- $\mathcal{L}$: set of compartments, indexed by l.
- $\mathcal{T} = \{1, \dots, T\}$: set of planning periods, indexed by t.

Let $n = |\mathcal{V}'|$.

A.3 Parameters

- d_i^{mt}: demand/consumption of product m at customer i during period t.
- I_i^{m0}: initial inventory of product m at customer i.
- C_i^m: capacity of the tank dedicated to product m at customer i.
- Q_{lk}: capacity of compartment l of vehicle k.
- c_{ij}: routing cost of travelling directly from node i to node j; for the benchmark data this is normally symmetric.
- h_i^{mt}: unit holding cost of product m at customer i at the end of period t.

The supplier is assumed to have sufficient product availability to cover the planning horizon.

A.4 Decision variables

- $I_i^{mt} \geq 0$: inventory of product m at customer i at the end of period t.

- $q_i^{mlkt} \geq 0$: quantity of product m delivered to customer i from compartment l of vehicle k in period t.
- $x_{ij}^{kt} \in \{0,1\}$: 1 if vehicle k travels directly from node i to node j in period t, 0 otherwise, for $i \neq j$.
- $y_i^{kt} \in \{0,1\}$: 1 if vehicle k visits customer i in period t; $y_0^{kt} = 1$ means that vehicle k is used in period t.
- $w^{mlkt} \in \{0,1\}$: 1 if product m is loaded in compartment l of vehicle k in period t.
- $z_i^{mlkt} \in \{0,1\}$: 1 if compartment l of vehicle k delivers product m to customer i in period t.
- $u_i^{kt} \geq 0$: MTZ order variable for customer i on the route of vehicle k in period t.

A.5 Objective function

A0 — Minimize inventory holding and routing costs

$$\min \sum_{i \in \mathcal{V}'} \sum_{m \in \mathcal{M}} \sum_{t \in \mathcal{T}} h_i^{mt} I_i^{mt} + \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} \sum_{\substack{i,j \in \mathcal{V} \\ i \neq j}} c_{ij} x_{ij}^{kt}.$$

This is the benchmark-oriented objective. The first term penalizes inventory carried at customer sites; the second term is the total routing cost. When validating against published benchmark values, the same objective coefficients and assumptions as the reference instance must be used.

A.6 Inventory constraints

A1 — Inventory balance

$$I_i^{mt} = I_i^{m,t-1} + \sum_{l \in \mathcal{L}} \sum_{k \in \mathcal{K}} q_i^{mlkt} - d_i^{mt}, \qquad i \in \mathcal{V}',\ m \in \mathcal{M},\ t \in \mathcal{T}.$$

The end-of-period inventory equals the previous inventory plus all deliveries received during the period minus the period demand.

A2 — End-of-period tank capacity

$$0 \leq I_i^{mt} \leq C_i^m, \qquad i \in \mathcal{V}',\ m \in \mathcal{M},\ t \in \mathcal{T}.$$

Inventory cannot be negative and cannot exceed the capacity of the corresponding customer tank.

A3 — Available tank space before consumption

$$\sum_{l \in \mathcal{L}} \sum_{k \in \mathcal{K}} q_i^{mlkt} \leq C_i^m - I_i^{m,t-1}, \qquad i \in \mathcal{V}',\ m \in \mathcal{M},\ t \in \mathcal{T}.$$

The total quantity delivered during period t cannot exceed the free space in the tank at the beginning of the period. This prevents a temporary tank overflow even if later consumption would make the end-of-period inventory feasible.

A.7 Delivery and compartment constraints

A4 — Delivery by a vehicle requires a customer visit

$$\sum_{l \in \mathcal{L}} q_i^{mlkt} \leq C_i^m y_i^{kt}, \qquad i \in \mathcal{V}',\ m \in \mathcal{M},\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

If vehicle k does not visit customer i in period t, it cannot deliver any quantity of product m to that customer.

A5 — Quantity can be delivered only from an assigned compartment

$$q_i^{mlkt} \leq Q_{lk} z_i^{mlkt}, \qquad i \in \mathcal{V}',\ m \in \mathcal{M},\ l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

If $z_i^{mlkt} = 0$, the delivered quantity is zero. If the compartment is assigned to customer i and product m, the delivery cannot exceed its capacity.

A6 — Compartment capacity

$$\sum_{i \in \mathcal{V}'} q_i^{mlkt} \leq Q_{lk}, \qquad m \in \mathcal{M},\ l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

For any product-compartment-vehicle-period combination, the total quantity discharged from the compartment cannot exceed its physical capacity.

A7 — Assignment to a customer requires a visit

$$z_i^{mlkt} \leq y_i^{kt}, \qquad i \in \mathcal{V}',\ m \in \mathcal{M},\ l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

A compartment cannot be assigned to customer i unless vehicle k visits that customer in the same period.

A8 — Assignment requires the corresponding product to be loaded

$$z_i^{mlkt} \leq w^{mlkt}, \qquad i \in \mathcal{V}',\ m \in \mathcal{M},\ l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

A compartment can deliver product m only if product m is the product loaded in that compartment.

A9 — At most one product per compartment

$$\sum_{m \in \mathcal{M}} w^{mlkt} \leq 1, \qquad l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

Products cannot be mixed inside one compartment.

A10 — Full-compartment delivery (unsplit compartments)

$$q_i^{mlkt} \geq Q_{lk} z_i^{mlkt}, \qquad i \in \mathcal{V}',\ m \in \mathcal{M},\ l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

Combined with A5, this gives $q_i^{mlkt} = Q_{lk} z_i^{mlkt}$. Hence, whenever a compartment is assigned to a customer tank, its full content is delivered. This is the defining feature of the **Unsplit Compartments** case.

A11 — One customer/product assignment per compartment (useful strengthening)

$$\sum_{i \in \mathcal{V}'} \sum_{m \in \mathcal{M}} z_i^{mlkt} \leq 1, \qquad l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

This explicitly states that a compartment is assigned to at most one customer and one product during the route. In the full-load case this condition is already implied by the capacity and product-assignment logic, but adding it is a useful strengthening and corresponds to inequality (58) in Coelho and Laporte.

A12 — A visited customer must receive at least one compartment assignment (strengthening)

$$\sum_{m \in \mathcal{M}} \sum_{l \in \mathcal{L}} z_i^{mlkt} \geq y_i^{kt}, \qquad i \in \mathcal{V}',\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

A vehicle should not visit a customer without making a delivery. This valid inequality removes meaningless visits and strengthens the formulation.

A.8 Directed routing constraints

A13 — Exactly one outgoing arc from every visited customer

$$\sum_{\substack{j \in \mathcal{V} \\ j \neq i}} x_{ij}^{kt} = y_i^{kt}, \qquad i \in \mathcal{V}',\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

If customer i is visited by vehicle k, exactly one arc leaves i on that route; otherwise no outgoing arc is used.

A14 — Exactly one incoming arc to every visited customer

$$\sum_{\substack{j \in \mathcal{V} \\ j \neq i}} x_{ji}^{kt} = y_i^{kt}, \qquad i \in \mathcal{V}',\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

If customer i is visited, exactly one route arc enters it. Together, A13 and A14 imply that the same vehicle visits a customer at most once in a period.

A15 — Route departure from the depot

$$\sum_{j \in \mathcal{V}'} x_{0j}^{kt} = y_0^{kt}, \qquad k \in \mathcal{K},\ t \in \mathcal{T}.$$

If vehicle k is used in period t, it leaves the depot exactly once. If it is not used, no route leaves the depot.

A16 — Route return to the depot

$$\sum_{i \in \mathcal{V}'} x_{i0}^{kt} = y_0^{kt}, \qquad k \in \mathcal{K},\ t \in \mathcal{T}.$$

Every used vehicle returns to the depot exactly once in the period.

A17 — A customer can be visited only by an active vehicle

$$y_i^{kt} \le y_0^{kt}, \qquad i \in \mathcal{V}',\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

The depot-use variable activates the complete route. A vehicle cannot serve a customer if it does not perform a route in that period.

A.9 MTZ subtour-elimination constraints

The original Coelho–Laporte formulation uses an exponential family of subset subtour-elimination constraints and separates them dynamically in a branch-and-cut algorithm. For a standard CPLEX/Gurobi implementation, we replace that block by the following compact MTZ formulation.

A18 — MTZ order-variable bounds

$$y_i^{kt} \le u_i^{kt} \le n\, y_i^{kt}, \qquad i \in \mathcal{V}',\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

An unvisited customer has order zero. A visited customer receives an order between 1 and n.

A19 — MTZ ordering

$$u_i^{kt} - u_j^{kt} + n\, x_{ij}^{kt} \le n - y_j^{kt}, \qquad i,j \in \mathcal{V}',\ i \ne j,\ k \in \mathcal{K},\ t \in \mathcal{T}.$$

If arc $i \to j$ is used, then both customers are visited and the constraint reduces to $u_j^{kt} \ge u_i^{kt} + 1$. Therefore a directed cycle containing only customers is impossible. This replaces the exponential subtour-elimination family used in the original paper.

A.10 Variable domains

A20 — Domains

$$x_{ij}^{kt},\ y_i^{kt},\ w^{mlkt},\ z_i^{mlkt} \in \{0,1\},$$

$$q_i^{mlkt} \ge 0, \qquad I_i^{mt} \ge 0, \qquad u_i^{kt} \ge 0.$$

All directed arc variables are binary. In particular, unlike the original undirected formulation, no routing variable needs the value 2: a one-customer route is represented by $a_{0i}^{kt} = a_{i0}^{kt} = 1$.

A.11 Meaning of "split tanks" in this formulation

There is deliberately **no** constraint of the form

$$\sum_{k \in \mathcal{K}} y_i^{kt} \leq 1.$$

Hence different vehicles may visit the same customer in the same period. More specifically, a customer tank may receive deliveries from several vehicles, which is the *split-tank* assumption. However, because y_i^{kt} is binary and A13–A14 impose degree one for each vehicle, a given vehicle visits a given customer at most once during the period.